

Übungsprotokoll

INSY– Informationssysteme

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	Abgabedatum: 19/03/2026	Gruppe: INSY2	Note:
Leitung: Dipl.-Ing.(FH) MESTL Alexander	Mitübende:		
Übungsbezeichnung: XML (III) - XSLT-Übungsaufgaben			

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1 Aufgabenstellung

Skriptum insyExercise3.pdf:

Seite 40, 1. Beispiel "teams.xml" und Seite 41, 3. Beispiel "positions.xml" sowie Seite 44, 1. Beispiel "teams.xml" und Seite 45, 3. Beispiel "positions.xml" - jeweils die gleiche Ausgabedatei, einmal deklarativ und einmal prozedura

2 Abstract (English)

This exercise focuses on transforming XML data using XSLT.

The goal was to generate identical XML output files using two different approaches: declarative (using templates and apply-templates) and procedural (using for-each loops).

The tasks included generating teams.xml and positions.xml from a given input file (euro.xml).

The exercise demonstrates the difference between rule-based and step-by-step processing in XSLT.

3 Theoretische Grundlagen

XSLT (Extensible Stylesheet Language Transformations) wird verwendet, um XML-Dokumente in andere Formate (z. B. XML, HTML) zu transformieren.

Es gibt zwei grundlegende Vorgehensweisen:

Deklarative Programmierung

- Verwendung von `<xsl:template>` und `<xsl:apply-templates>`
- Verarbeitung erfolgt automatisch über Regeln
- Modular und übersichtlich

Prozedurale Programmierung

- Verwendung von `<xsl:for-each>`
- Ablauf wird explizit gesteuert
- Ähnlich wie klassische Programmierung

4 Übungsdurchführung

teams_deklarativ.xsl

```
<?xml version="1.0" encoding="UTF-8"?>
<xsl:stylesheet xmlns:xsl="http://www.w3.org/1999/XSL/Transform"
version="2.0">

  <xsl:output method="xml" indent="yes"/>

  <xsl:template match="/">
    <teams>
      <xsl:apply-templates select="euro/teams/team"/>
    </teams>
  </xsl:template>

  <xsl:template match="team">
    <team>
      <xsl:attribute name="code">
        <xsl:value-of select="nation/short-name"/>
      </xsl:attribute>
      <xsl:apply-templates select="position/player"/>
    </team>
  </xsl:template>

  <xsl:template match="player">
    <player>
      <xsl:value-of select="concat((name|pl)/first-name, ' ',
(name|pl)/last-name)"/>
    </player>
  </xsl:template>

</xsl:stylesheet>
```

positions deklarativ.xsl

```
<?xml version="1.0" encoding="UTF-8"?>
<xsl:stylesheet xmlns:xsl="http://www.w3.org/1999/XSL/Transform"
version="2.0">

    <xsl:output method="xml" indent="yes"/>

    <xsl:param name="defense" select="'defense'"/>
    <xsl:param name="playground" select="'playground'"/>
    <xsl:param name="attack" select="'attack'"/>

    <xsl:template match="/">
        <positions>
            <position type="{ $defense }">
                <xsl:apply-templates select="euro/teams/team">
                    <xsl:with-param name="ptype" select="$defense"/>
                </xsl:apply-templates>
            </position>

            <position type="{ $playground }">
                <xsl:apply-templates select="euro/teams/team">
                    <xsl:with-param name="ptype"
select="$playground"/>
                </xsl:apply-templates>
            </position>

            <position type="{ $attack }">
                <xsl:apply-templates select="euro/teams/team">
                    <xsl:with-param name="ptype" select="$attack"/>
                </xsl:apply-templates>
            </position>
        </positions>
    </xsl:template>

    <xsl:template match="team">
        <xsl:param name="ptype"/>
        <team>
            <xsl:attribute name="code">
                <xsl:value-of select="nation/short-name"/>
            </xsl:attribute>
            <xsl:apply-templates select="position[@type =
$ptype]/player"/>
        </team>
    </xsl:template>
```

```
<xsl:template match="player">
  <player>
    <xsl:value-of select="concat((name|pl)/first-name, ' ',
(name|pl)/last-name)"/>
  </player>
</xsl:template>

</xsl:stylesheet>
```

teams_prozedural.xsl

```
<?xml version="1.0" encoding="UTF-8"?>
<xsl:stylesheet xmlns:xsl="http://www.w3.org/1999/XSL/Transform"
version="2.0">

  <xsl:output method="xml" indent="yes"/>

  <xsl:template match="/">
    <teams>
      <xsl:for-each select="euro/teams/team">
        <team>
          <xsl:attribute name="code">
            <xsl:value-of select="nation/short-name"/>
          </xsl:attribute>

          <xsl:for-each select="position/player">
            <player>
              <xsl:value-of
select="concat((name|pl)/first-name, ' ', (name|pl)/last-name)"/>
            </player>
          </xsl:for-each>
        </team>
      </xsl:for-each>
    </teams>
  </xsl:template>

</xsl:stylesheet>
```

positions_prozedural.xsl

```
<?xml version="1.0" encoding="UTF-8"?>
<xsl:stylesheet xmlns:xsl="http://www.w3.org/1999/XSL/Transform"
version="2.0">

  <xsl:output method="xml" indent="yes"/>

  <xsl:variable name="defense" select="'defense'"/>
  <xsl:variable name="playground" select="'playground'"/>
  <xsl:variable name="attack" select="'attack'"/>

  <xsl:template match="/">
    <positions>
      <position type="{ $defense }">
        <xsl:for-each select="euro/teams/team">
          <team>
            <xsl:attribute name="code">
              <xsl:value-of select="nation/short-
name"/>
            </xsl:attribute>

            <xsl:for-each select="position[@type =
$defense]/player">
              <player>
                <xsl:value-of
select="concat((name|pl)/first-name, ' ', (name|pl)/last-name)"/>
              </player>
            </xsl:for-each>
          </team>
        </xsl:for-each>
      </position>

      <position type="{ $playground }">
        <xsl:for-each select="euro/teams/team">
          <team>
            <xsl:attribute name="code">
              <xsl:value-of select="nation/short-
name"/>
            </xsl:attribute>

            <xsl:for-each select="position[@type =
$playground]/player">
              <player>
                <xsl:value-of
```

```
select="concat((name|pl)/first-name, ' ', (name|pl)/last-name)"/>
    </player>
  </xsl:for-each>
</team>
</xsl:for-each>
</position>

<position type="{ $attack }">
  <xsl:for-each select="euro/teams/team">
    <team>
      <xsl:attribute name="code">
        <xsl:value-of select="nation/short-
name"/>
      </xsl:attribute>

      <xsl:for-each select="position[@type =
$attack]/player">
        <player>
          <xsl:value-of
select="concat((name|pl)/first-name, ' ', (name|pl)/last-name)"/>
        </player>
      </xsl:for-each>
    </team>
  </xsl:for-each>
</position>
</positions>
</xsl:template>

</xsl:stylesheet>
```

5 Ergebnisse

Teams.xml

```
<?xml version="1.0" encoding="UTF-8"?><teams>
  <team code="FRA">
    <player>Laurent Koscielny</player>
    <player>Djibril Sidibe</player>
    <player>Antoine Griezmann</player>
    <player>Olivier Giroud</player>
    <player>Nabil Fekir</player>
    <player>Florian Thauvin</player>
    <player>Alexandre Lacazette</player>
  </team>
  <team code="SPA">
    <player>Jordi Alba</player>
    <player>Dani Carvajal</player>
    <player>Sergio Ramos</player>
    <player>Alberto Moreno</player>
    <player>Marc Bartra</player>
    <player>Marco Asensio</player>
    <player>Andres Iniesta</player>
    <player>David Silva</player>
    <player>Javi Martinez</player>
    <player>Ander Herrera</player>
    <player>David Villa</player>
    <player>Diego Costa</player>
    <player>Lucas Vazquez</player>
    <player>Aritz Aduriz</player>
  </team>
  <team code="GER">
```

Position.xml

```
<?xml version="1.0" encoding="UTF-8"?><positions>
  ⚡ <position type="defense">
    <team code="FRA">
      <player>Laurent Koscielny</player>
      <player>Djibril Sidibe</player>
    </team>
    <team code="SPA">
      <player>Jordi Alba</player>
      <player>Dani Carvajal</player>
      <player>Sergio Ramos</player>
      <player>Alberto Moreno</player>
      <player>Marc Bartra</player>
    </team>
    <team code="GER">
      <player>Jerome Boateng</player>
      <player>Marcel Halstenberg</player>
      <player>Mats Hummels</player>
      <player>Niklas Süle</player>
    </team>
    <team code="GB">
      <player>Ryan Bertrand</player>
      <player>Joe Gomez</player>
      <player>Phil Jones</player>
    </team>
    <team code="POR">
      <player>Luis Neto</player>
      <player>Ricardo Pereira</player>
    </team>
  </position>
</positions>
```

6 Code

7 Kommentar